The numbers with exponential generating function $(e^x - 1)/\cos 2x$, modulo 4 and 5

A proof of two conjectures in OEIS A380056

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Abstract

The OEIS entry A380056, contributed by Paul D. Hanna, lists the numbers $a(n)$ defined by the exponential generating function $(e^x - 1)/\cos 2x$; that is, $a(n)/n!$ is the coefficient of x^n in the power series of this function. The first terms are $1, 1, 13, 25, 441, 1261, \ldots$. Hanna conjectured that $a(n)$ leaves remainder 1 on division by 4 for every $n \geq 1$, and that $a(4n)$ is divisible by 5. We prove both, and more: modulo 5, $a(n) \equiv 2 \cdot 2^n + 3 \cdot 4^n$, so the remainders repeat in the cycle $1, 1, 3, 0$, and $a(n)$ is divisible by 5 exactly when n is a multiple of 4. The proof first rewrites the definition as a recurrence that uses only integers, and then solves this recurrence modulo 4 and modulo 5. Modulo 5 the solution is a combination of two geometric sequences. The integer exponential coefficients $n![x^n]\cos 2x$ agree modulo 5 with $n![x^n]\,(e^x + e^{-x})/2$. The results have also been checked by computer and proved formally in the Lean proof assistant.

How this paper was produced. This paper reports the result of an AI research effort directed by Dev D. Goyal. ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning and working autonomously from prompts by Dev D. Goyal, found the proof, wrote its Lean 4 formalization, and wrote a first draft of the paper. Dev D. Goyal chose the problems and asked for proofs, formal verification and revisions, but did not supply proof ideas. Claude Opus 5.5, used with Claude Code, then checked the work independently: it rebuilt the formal proofs from source and confirmed that they rely only on Lean's three standard axioms, recomputed the sequence from its definition compared it with the published terms and tested every claim numerically, and read every step of the proofs. It also wrote the present version of this paper. Dev D. Goyal has reviewed the proofs and is responsible for them; this review is not an independent expert validation. The paper has not been peer reviewed. The section "About this paper" at the end gives the details.

1 Introduction

1.1 The sequence

The *On-Line Encyclopedia of Integer Sequences* (OEIS) [3] is a large public database of integer sequences. Many entries also record *conjectures*: patterns that the contributor observed in the data but did not prove.

This paper is about entry A380056 [1], contributed by Paul D. Hanna in January 2025. Its name is:

```
E.g.f.  (exp(x) - 1)/cos(2*x).
```

— OEIS A380056, Paul D. Hanna, January 2025

"E.g.f." stands for *exponential generating function* (Section 2). The name means that

$$\sum_{n\geq 0} a(n)\,\frac{x^n}{n!} = \frac{e^x - 1}{\cos 2x}, \tag{1}$$

where $e^x = 1 + x + \frac{x^2}{2!} + \cdots$ and $\cos 2x = 1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \cdots$ are the usual power series. In other words, expand $(e^x - 1)/\cos 2x$ as a power series; the coefficient of x^n, multiplied by $n!$, is $a(n)$. The constant term is 0, so $a(0) = 0$; the entry starts at $n = 1$. The first terms, from $n = 1$, are 1, 1, 13, 25, 441, 1261, 30213, 115025, 3529201, 16792021, ... It is not obvious from the definition that these numbers are integers; Lemma 4.1 below shows that they are.

1.2 The question

The entry contains two conjectures.

> `Conjecture: a(n) == 1 (mod 4) for n >= 1.`
>
> `Conjecture: a(4*n) is divisible by 5 for n >= 1.`
>
> — OEIS A380056, Paul D. Hanna, January 2025

Here `a(n) == 1 (mod 4)` means that $a(n)$ leaves remainder 1 on division by 4. The table shows the remainders of the first terms modulo 4 and modulo 5. The indices in bold are the multiples of 4.

n	1	2	3	**4**	5	6	7	**8**	9	10	11	**12**	13	14	15	**16**
$a(n)$ mod 4	1	1	1	**1**	1	1	1	**1**	1	1	1	**1**	1	1	1	**1**
$a(n)$ mod 5	1	1	3	**0**	1	1	3	**0**	1	1	3	**0**	1	1	3	**0**

n	17	18	19	**20**	21	22	23	**24**	25	26	27	**28**	29	30	31	**32**
$a(n)$ mod 4	1	1	1	**1**	1	1	1	**1**	1	1	1	**1**	1	1	1	**1**
$a(n)$ mod 5	1	1	3	**0**	1	1	3	**0**	1	1	3	**0**	1	1	3	**0**

Modulo 5 the data show more than the conjecture: the remainders repeat in the cycle $1, 1, 3, 0$.

1.3 What this paper proves

Main result (Theorem 3.1). *The numbers $a(n)$ are integers, and*

- $a(n) \equiv 1 \pmod 4$ *for every $n \geq 1$;*
- $a(n) \equiv 2 \cdot 2^n + 3 \cdot 4^n \pmod 5$ *for every $n \geq 0$. So the remainder of $a(n)$ modulo 5 is $0, 1, 1, 3$ according as n leaves remainder $0, 1, 2, 3$ on division by 4; in particular $a(n)$ is divisible by 5 exactly when n is a multiple of 4.*

Both conjectures follow (Corollary 3.2).

1.4 What was known

The entry records, among other formulas, an asymptotic formula due to Vaclav Kotesovec and two expressions for $a(n)$ (formulas (3.a) and (3.b) in its list of formulas) in terms of the

Euler (secant) numbers $S_0, S_1, S_2, \cdots = 1, 1, 5, 61, 1385, \ldots$ (OEIS A000364 [2]):

$$a(2n+1) = \sum_{k=0}^{n} \binom{2n+1}{2k+1} S_{n-k}\, 4^{n-k} \quad (n \geq 0), \qquad a(2n) = \sum_{k=1}^{n} \binom{2n}{2k} S_{n-k}\, 4^{n-k} \quad (n \geq 1).$$

These expressions show that the $a(n)$ are integers. They also imply the first conjecture directly: in each sum, every term except the one with $k = n$ contains a factor 4, and the term with $k = n$ is 1. So the first conjecture already follows from formulas recorded in the entry. We are not aware of an earlier proof of the second conjecture, or of the formula modulo 5.

1.5 Motivation

Exponential generating functions are the natural way to encode sequences that count arrangements of labelled objects, and they appear throughout combinatorics. Their coefficients $n! \cdot [x^n]\, F$ are often integers, even though the series F itself has fractional coefficients, and the remainders of such integers modulo small numbers often follow patterns. The difficulty in proving such patterns is that the factorials in the denominators prevent a direct reduction modulo m. Lemma 2.1 gives a way around this, and the proof of the result modulo 5 gives an example where, modulo 5, a trigonometric function turns into a combination of exponentials.

1.6 How to read this paper

Section 2 explains the notions we need: power series, remainders, exponential generating functions, and a way to multiply them without fractions. Section 3 states the results precisely, and Section 4 proves them. Section 5 describes how the results were checked, by computer and by a formal proof in Lean. Section 6 discusses other moduli.

2 Background

This section collects the definitions and facts used later. Readers who know formal power series and congruences can skip it and return when needed.

2.1 Formal power series

A *formal power series* is an expression

$$U(x) = u_0 + u_1 x + u_2 x^2 + u_3 x^3 + \cdots,$$

determined by its sequence of coefficients $u_0, u_1, u_2, \ldots$; see [5, 4]. No number is ever substituted for x, so questions of convergence do not arise. We write $[x^n]\, U = u_n$ for the coefficient of x^n in U; two series are equal when all their coefficients are equal.

Series are added coefficient by coefficient and multiplied like polynomials:

$$[x^n]\,(UV) = u_0 v_n + u_1 v_{n-1} + \cdots + u_n v_0.$$

Each coefficient of a product is a finite sum. For example, $(1-x)(1+x+x^2+x^3+\cdots) = 1$, so $1 + x + x^2 + \cdots$ is the *multiplicative inverse* of $1 - x$, written $\frac{1}{1-x}$. More generally, every series D with constant term 1 has a unique multiplicative inverse $\frac{1}{D}$, whose coefficients are determined one at a time by $D \cdot \frac{1}{D} = 1$ using only addition, subtraction and multiplication.

Whenever a quotient of power series appears in this paper, its denominator has constant term 1, so the quotient is again a power series.

A *commutative ring* is a set with an addition, a subtraction and a multiplication that satisfy the usual rules of arithmetic; examples are $\mathbb{Z}$, $\mathbb{Q}$ and the integers modulo m. For a commutative ring R we write $R[[x]]$ for the set of power series with coefficients in R; for instance, writing $\mathbb{F}_2$ for the integers modulo 2, $\mathbb{F}_2[[x]]$ consists of the series with coefficients modulo 2. We write $U(0)$ for the constant term u_0 of U, and $U = O(x^d)$ when the coefficients of $1, x, \ldots, x^{d-1}$ in U are all zero. Thus $U - V = O(x^d)$ means that U and V have the same coefficient of x^n for every $n < d$. A series with constant term 0 and coefficient of x equal to 1, that is, a series of the form $x + u_2x^2 + u_3x^3 + \cdots$, is called *normalized*.

2.2 Remainders and congruences

For an integer $m \geq 2$ we write $a \equiv b \pmod{m}$ ("a is congruent to b modulo m") when $a - b$ is divisible by m, that is, when a and b leave the same remainder after division by m. The remainder of a on division by m is called its *residue* modulo m; it is one of $0, 1, \ldots, m-1$.

Addition and multiplication are compatible with congruences, so the residues $0, 1, \ldots, m-1$, added and multiplied modulo m, form a ring, written $\mathbb{Z}/m\mathbb{Z}$. When $m = p$ is a prime we also write $\mathbb{F}_p$. In $\mathbb{F}_2 = \{0, 1\}$ we have $1 + 1 = 0$, so addition and subtraction coincide.

For power series with integer coefficients, $U \equiv V \pmod{m}$ means that *every* coefficient of $U - V$ is divisible by m. Equivalently, the two series become equal when every coefficient is replaced by its residue; we call this *reducing* the series modulo m. The key fact is the following.

Reduction modulo m is compatible with all the operations above. To reduce a sum, a product, a substitution, or an inverse, we may reduce the ingredients first and then compute in $\mathbb{Z}/m\mathbb{Z}$. This holds because every coefficient of the result is obtained from finitely many coefficients of the ingredients by adding, subtracting and multiplying, and these operations are compatible with remainders.

Rings. Some statements below hold over every commutative ring, for example $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{Z}/m\mathbb{Z}$. No division is used in them; this matters because in $\mathbb{Z}/m\mathbb{Z}$ division is not always possible (for instance, 2 has no inverse in $\mathbb{Z}/4\mathbb{Z}$).

2.3 Exponential generating functions

The *exponential generating function* (e.g.f.) of a sequence $f(0), f(1), f(2), \ldots$ is the power series

$$\mathcal{E}(f) = \sum_{n \geq 0} f(n) \, \frac{x^n}{n!}.$$

The division by $n!$ is a convention that suits sequences which count arrangements of n labelled objects. Two examples matter here. The constant sequence $1, 1, 1, \ldots$ has e.g.f. $\sum_n x^n/n! = e^x$, the exponential series. And the sequence $1, 0, -1, 0, 1, 0, \ldots$ has e.g.f. $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots = \cos x$. We use these only as formal power series: e^x and $\cos x$ are defined by these series, and x is never given a value.

Replacing x by $2x$ multiplies the coefficient of x^n by 2^n. So

$$\cos 2x = \sum_{j \geq 0} (-1)^j \frac{(2x)^{2j}}{(2j)!} = \sum_{j \geq 0} (-4)^j \, \frac{x^{2j}}{(2j)!},$$

that is, $\cos 2x$ is the e.g.f. of the sequence c with $c(2j) = (-4)^j$ and $c(2j+1) = 0$: $c = 1, 0, -4, 0, 16, 0, -64, \ldots$. Its constant term is 1, so $\cos 2x$ has a multiplicative inverse among the power series with rational coefficients, and the quotient $(e^x - 1)/\cos 2x$ is a power series.

2.4 Multiplying e.g.f.s without fractions

Lemma 2.1 (Binomial convolution)**.** *For two sequences f and g, define the sequence $f \star g$ by*

$$(f \star g)(n) = \sum_{k=0}^{n} \binom{n}{k} f(k)\, g(n-k).$$

Then $\mathcal{E}(f) \cdot \mathcal{E}(g) = \mathcal{E}(f \star g)$.

Proof. The coefficient of x^n in the product is $\sum_k \frac{f(k)}{k!} \cdot \frac{g(n-k)}{(n-k)!}$. Multiply by $n!$ and use $\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$. □

The point of this lemma is that $f \star g$ is computed from f and g with additions and multiplications of whole numbers only. So it makes sense for sequences of residues modulo any m, and reduction modulo m is compatible with it. The e.g.f. itself, with its denominators $n!$, cannot be reduced modulo m directly.

Two simple facts about $\star$ will be used. First, the sequence $\delta = 1, 0, 0, 0, \ldots$ is neutral: $\delta \star f = f$ for every f. Second, write g_s for the geometric sequence $g_s(n) = s^n$ (with $s^0 = 1$), whose e.g.f. is e^{sx}. Then

$$g_s \star g_t = g_{s+t},$$

because $\sum_k \binom{n}{k} s^k t^{n-k} = (s+t)^n$ by the binomial theorem. (In e.g.f. language this is $e^{sx} e^{tx} = e^{(s+t)x}$.) These facts hold for numbers s, t in any commutative ring, in particular for residues modulo 5.

3 Results

Theorem 3.1. *The numbers $a(n)$ defined by* (1) *are integers, and:*

(a) $a(n) \equiv 1 \pmod 4$ *for every $n \geq 1$;*

(b) $a(n) \equiv 2 \cdot 2^n + 3 \cdot 4^n \pmod 5$ *for every $n \geq 0$. Equivalently, $a(n) \bmod 5$ is $0, 1, 1, 3$ according as $n \equiv 0, 1, 2, 3 \pmod 4$. In particular, 5 divides $a(n)$ if and only if 4 divides n.*

Corollary 3.2 (Hanna's conjectures)**.** $a(n) \equiv 1 \pmod 4$ *for $n \geq 1$, and $a(4n)$ is divisible by 5 for every $n \geq 1$.*

Proof. The first statement is part (a). The second is the case "4 divides the index" of part (b). □

4 Proofs

The idea in brief. By Lemma 2.1, multiplying (1) by $\cos 2x$ turns the definition into a recurrence for $a(n)$ that uses only integers. Put $c(n) = n![x^n] \cos 2x$. Every $c(n)$ with $n > 0$ is

divisible by 4, so the recurrence gives $a(n) \equiv 1 \pmod 4$ for $n \geq 1$. Modulo 5, we have $-4 \equiv 1$. Thus $c(n)$ agrees with $n![x^n]\,(e^x + e^{-x})/2$ modulo 5. The integer exponential coefficients of e^{sx} form the geometric sequence g_s. Binomial convolution of these sequences gives the result.

Throughout, let c be the coefficient sequence of $\cos 2x$ (Section 2), and let $d = 0, 1, 1, 1, \ldots$ be that of $e^x - 1$, so that $\mathcal{E}(c) = \cos 2x$ and $\mathcal{E}(d) = e^x - 1$.

Lemma 4.1. *In every commutative ring there is exactly one sequence a with $a \star c = d$, namely the one given by $a(0) = 0$ and*

$$a(n) = 1 - \sum_{k=0}^{n-1} \binom{n}{k} a(k)\, c(n-k) \qquad (n \geq 1).$$

Over the integers this is the sequence of OEIS A380056. In particular its terms are integers, and reducing it modulo m gives the solution of $a \star c = d$ over $\mathbb{Z}/m\mathbb{Z}$.

Proof. Since $c(0) = 1$, the equation $(a \star c)(n) = d(n)$ reads $a(n) + \sum_{k<n} \binom{n}{k} a(k)c(n-k) = d(n)$, which is the displayed recurrence (with $d(0) = 0$ and $d(n) = 1$ for $n \geq 1$). It determines each $a(n)$ from the earlier terms by additions, subtractions and multiplications, so it has exactly one solution in every commutative ring, and reduction modulo m maps one solution to the other. Over the rational numbers, Lemma 2.1 gives $\mathcal{E}(a) \cdot \cos 2x = \mathcal{E}(a \star c) = e^x - 1$, and dividing by $\cos 2x$ shows that $\mathcal{E}(a)$ is the series in (1). So the integer solution is the sequence of the entry. □

Proof of Theorem 3.1. The integrality was shown in Lemma 4.1.

(a) Modulo 4, every term $c(2j) = (-4)^j$ with $j \geq 1$ is 0, and the odd terms are 0 anyway. So $c \equiv \delta \pmod 4$, the neutral sequence $1, 0, 0, \ldots$. The sequence d solves $d \star \delta = d$, so by the uniqueness in Lemma 4.1, applied over $\mathbb{Z}/4\mathbb{Z}$, we get $a \equiv d \pmod 4$. That is, $a(n) \equiv 1 \pmod 4$ for every $n \geq 1$.

(b) Work in $\mathbb{F}_5$, the residues modulo 5. There $-4 = 1$, so $c(n) = 1$ for even n and $c(n) = 0$ for odd n. In terms of geometric sequences,

$$c = \tfrac{1}{2}(g_1 + g_{-1}), \qquad d = g_1 - g_0,$$

because $\frac{1}{2}(1^n + (-1)^n)$ is 1 for even n and 0 for odd n. (In $\mathbb{F}_5$ we can divide by 2: $\frac{1}{2} = 3$, since $2 \cdot 3 = 6 \equiv 1$.) We look for the solution as a combination of two geometric sequences. By $g_s \star g_t = g_{s+t}$, each geometric sequence satisfies $g_u \star c = \frac{1}{2}(g_{u+1} + g_{u-1})$. The choice $u = 2$ produces the term g_1 of d, and the choice $u = -1$ produces the term g_0; the factors 2 and -2 give these terms the coefficients 1 and -1 that they have in d. So we try

$$b = 2g_2 - 2g_{-1}.$$

Then

$$b \star c = 2 \cdot \tfrac{1}{2}(g_{2+1} + g_{2-1}) - 2 \cdot \tfrac{1}{2}(g_{-1+1} + g_{-1-1}) = g_3 + g_1 - g_0 - g_{-2}.$$

Since $-2 \equiv 3 \pmod 5$, the sequences g_{-2} and g_3 are equal, so the two remaining terms cancel and $b \star c = g_1 - g_0 = d$. By the uniqueness in Lemma 4.1, applied over $\mathbb{F}_5$, we get $a \equiv b$, that is,

$$a(n) \equiv 2 \cdot 2^n - 2 \cdot (-1)^n \equiv 2 \cdot 2^n + 3 \cdot 4^n \pmod 5,$$

since $-2 \equiv 3$ and $-1 \equiv 4$. Because $2^4 = 16 \equiv 1$ and $4^4 \equiv 1 \pmod 5$, the remainder depends only on n modulo 4. Evaluating at $n = 0, 1, 2, 3$ gives $2 + 3 = 5$, $4 + 12 = 16$, $8 + 48 = 56$ and $16 + 192 = 208$, whose remainders modulo 5 are $0, 1, 1, 3$. Only the first is 0, so 5 divides $a(n)$ exactly when 4 divides n. □

5 Checking the results

A mathematical proof can contain mistakes, so the results were checked in two independent ways.

Numerical check. We expanded $(e^x - 1)/\cos 2x$ as a power series with exact rational arithmetic. This reproduces all 22 terms printed in the OEIS entry. All $a(n)$ with $n < 400$ are integers and satisfy both parts of Theorem 3.1. The computation was done with the PARI/GP computer algebra system, using code that was written separately from the formal proofs (file `verification/independent_checks.gp` in the accompanying repository [8]). A computation can only test finitely many cases, so it is a check on the definitions and on the indexing, not a proof.

Formal proof in Lean. Lean [6] is a *proof assistant*: a program in which mathematical statements and their proofs are written in a precise formal language, and every step is checked mechanically down to the basic rules of logic. The theorems of Section 3 are proved in Lean 4 (version 4.35.0) on top of Mathlib [7], Lean's library of formalized mathematics. The statements are collected in the file `lean/Entries/A380056.lean` of the repository, in the namespace `OEIS.A380056`. There the series e^x and $\cos x$ are Mathlib's formal exponential and cosine series over $\mathbb{Q}$, with x replaced by $2x$ in the cosine. The integer sequence is defined by the recurrence of Lemma 4.1, and the proofs are in the file `lean/Families/ExpCosine.lean`. The table lists the results covered by the Lean formalization. The Lean formalization does not cover the additional arguments and observations in Section 6.

Statement in this paper	Lean theorem(s)
$a(n) = n!\,[x^n]\,A$, where $A \cdot \cos 2x = e^x - 1$ over $\mathbb{Q}$ and A is the only such series; the $a(n)$ are integers	`integer_egf_definition`, `source_egf`, `literal_source_equation`, `literal_source_unique`, `integer_coefficients`
Corollary 3.2: the two conjectures	`first_source_conjecture`, `second_source_conjecture`
Theorem 3.1(b) and its consequences	`mod_five_exponential_formula`, `exact_mod_five_pattern`, `exact_divisibility_by_five`, `mod_five_period`

The Lean project was rebuilt from source, and the axioms on which each of these theorems depends were listed with Lean's `#print axioms` command. Every one depends only on the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`); no step is left unproved. A formal proof certifies exactly the statement that Lean checks. Whether that statement says what the OEIS entry means is a separate question, so the Lean definitions were also compared with the definition in the OEIS entry (Section 1.1). The principal theorems are stated once more, using only Mathlib's definitions, in the file `lean/Challenges/ShortResults.lean`, and Lean's Comparator tool confirms that the proved theorems have exactly these statements.

6 Discussion

Scope. Theorem 3.1 describes the remainders of every $a(n)$ modulo 4 and modulo 5, so it settles both conjectures completely.

Other moduli. A computer calculation for $n < 400$ suggests that the remainders of $a(n)$ are eventually periodic for other moduli as well:

- modulo 3, with period 8 from $n = 1$ on;
- modulo 7, with period 48 from $n = 1$ on;
- modulo 8, with period 4 from $n = 3$ on (the remainders are $5, 1, 1, 5$ for $n \equiv 3, 0, 1, 2 \pmod 4$);
- modulo 16, with period 8 from $n = 3$ on.

We have not proved these observations. The proof of part (b) suggests where the periods $8 = 3^2 - 1$ and $48 = 7^2 - 1$ may come from. Modulo 5 the number -4 is a square ($-4 \equiv 1 = 1^2$), which is what made $\cos 2x$ a combination of exponentials. Modulo 3 and modulo 7 it is not a square, and a similar argument would need a larger ring in which -4 has a square root.

About this paper

How the work was done. This paper is one of 28 papers on conjectures from the OEIS that came out of an AI research effort directed by Dev D. Goyal [8]. The roles were as follows.

- *Dev D. Goyal* started the project, chose which problems to pursue, asked for proofs, formal verification and revisions, and decided what to publish. Dev D. Goyal did not supply proof strategies or intermediate arguments.
- *ChatGPT* (OpenAI), using the GPT-6 model with Pro reasoning, worked autonomously from those prompts. It searched the OEIS for suitable conjectures, found the proofs, wrote the Lean 4 formalization, and wrote a first draft of this paper.
- *Claude Opus 5.5*, used with Claude Code, checked the work independently. It rebuilt the Lean project from source and confirmed that the theorems listed in Section 5 rely only on Lean's standard axioms; compared the Lean definitions with the OEIS definition; recomputed the sequence with separately written PARI/GP code, compared it with the terms published in the OEIS, and tested every claim of the paper on many terms; and read every step of the proofs. It then wrote the present version of the paper, expanding the explanations and, in places, writing out an argument in more detail or by a more elementary route. The theorems of Section 3 and the proof strategies are ChatGPT's. The Discussion section, including the short arguments in it, was written by Claude Opus 5.5; its observations were checked by computer and are not part of the Lean formalization.

Dev D. Goyal has reviewed the proofs and is responsible for this paper; this review is not an independent expert validation.

Review status. This paper has not been peer reviewed, and it has not been submitted to a journal or to the OEIS.

Priority. An OEIS entry that labels a statement as a conjecture shows only that its contributor did not know a proof; it does not show that no proof existed. During the research only limited searches for earlier work were made. No claim of priority is made.

Corrections. Please report any error in the issue tracker of the repository [8], where the Lean sources and the numerical checks are also available.

Acknowledgements

The sequence and both conjectures are due to Paul D. Hanna [1]. The asymptotic formula mentioned in the introduction was added to the entry by Vaclav Kotesovec.

References

[1] P. D. Hanna, entry A380056 (E.g.f. $(e^x - 1)/\cos(2x)$), *The On-Line Encyclopedia of Integer Sequences*, January 2025. `https://oeis.org/A380056`. Text as of October 2026.

[2] N. J. A. Sloane, entry A000364 (Euler or secant numbers), *The On-Line Encyclopedia of Integer Sequences*. `https://oeis.org/A000364`.

[3] N. J. A. Sloane and The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*. `https://oeis.org`.

[4] R. P. Stanley, *Enumerative Combinatorics*, Volume 2, Cambridge University Press, 1999. (Chapter 5 treats formal power series, composition and compositional inverses.)

[5] I. Niven, Formal power series, *American Mathematical Monthly* 76 (1969), 871–889.

[6] L. de Moura and S. Ullrich, The Lean 4 theorem prover and programming language, in *Automated Deduction – CADE 28*, Lecture Notes in Computer Science 12699, Springer, 2021, 625–635.

[7] The mathlib Community, The Lean mathematical library, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, 367–381. Version used: commit `56a005855ea1`.

[8] D. D. Goyal, *Proofs and corrections of OEIS conjectures*, repository, 2026. `https://github.com/dvgyl/oeis-proofs`.